Real-Time Crisis Recognition and Response System for Protest Violence Detection

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***Abstract*— Public protests have become increasingly common as forms of civic expression, However, some quickly escalate from peaceful demonstrations into violent confrontations, causing threats to the people and requiring crisis management mechanisms. Conventional surveillance methods rely mostly on human operators, so there is poor scalability, errors due to exhaustion, and latencies. This paper proposes a real-time crisis recognition and response system specifically designed for automated violence detection in protest scenarios. The system integrates YOLO-based object detection for person and object localization, pose estimation for extracting 17-keypoint skeletal representations, and behavioral analysis that evaluates arm trajectories, motion velocity, and object proximity. Experimental evaluation on UCF Crime dataset achieved 93.4% classification accuracy with real-time inference at 28 FPS on GPU hardware, outperforming ConvLSTM baseline (89.7%) and YOLO-only approaches (91.2%). The proposed framework demonstrates the feasibility of multimodal AI for enhanced situational awareness in public safety applications while maintaining computational efficiency suitable for deployment.**

Keywords—Crisis detection, protest monitoring, violence recognition, real-time surveillance, computer vision, pose estimation, multimodal AI.

I. Introduction

Public protests are an essential part of democratic practice, allowing citizens to express political, social, and economic concerns. However, those gatherings can quickly escalate from a peaceful demonstration into a scene of violent clash, causing significant challenges for public safety authorities, police forces, as well as emergency services. Acts of violence can cause accidents, property destruction, as well as mass disruptions, hence the importance of fast detection and common response mechanisms.

Traditional surveillance and crisis response systems depend on social stream monitoring and CCTV feeds, which are limited by human operator tiredness, slow response times, and limited scalability. Additionally, recognizing between peaceful and violent or aggressive crowd activity frequently demands a precise understanding of behavioral indicators that are difficult for human observers to Detect in real time, such as throwing motions, fighting, or sudden changes in density.

The accomplishments in computer vision with regard to automatic action recognition and anomaly detection in video data are made possible by recent advances in AI and computer vision. However, the implementation of these techniques is difficult in high-risk, dynamic scenes such as public performances because they frequently lack multimodal integration, interpretability, and real-time capability.

In response, this study presents a Real-Time Crisis Recognition and Response System that detects violent individual or group activity in the course of protests and demonstrations. The system incorporates YOLO-based object detection, pose estimation for body joint localization, and behavioral analysis of the trajectories of the arms and assaultive motions in recognizing activities like throwdowns and fistfights. By including object distance analysis and multi-criteria score-based decision making on the basis of pose, velocity, and trajectory, the system accomplishes resilient classification of violent as opposed to non-violent activity. Observed incidents automatically create timestamped, geolocated alert messages with annotated keyframes that can guide law enforcers and emergency response units.

The main objectives are

1. A multimodal AI framework for per-person violence recognition in protest or demonstration areas that integrates object detection, pose estimation, person detection, and object interaction analysis.
2. An explainable behavioral scoring system which includes trajectory, velocity, and posture clues to enable clear and understandable decision-making.
3. A platform for real-time crisis alerts that improves operational situational awareness and lowers response latency for public safety organizations.

The rest of this paper is organized as follows: Section II discusses related work, Section III describes the methodology, Section IV presents experimental results, and Section V concludes the paper with discussion and future work

II. Related Work

Automated violence and abnormal-behavior identification from surveillance video is a rapidly developing field based on several threads of computer vision. Previous work falls into four overlaid themes related to this study: (1) appearance and motion-based violence Detection, (2) skeleton/pose-based and change-detection methods, (3) real-time systems and alarming, and (4) socio-legal and multimodal crisis detection (covering social-media signal processing).

A. Appearance- and motion-based violence recognition

In order to capture aggressive motion patterns, early and significant work in violence detection used spatiotemporal appearance features and optical flow. In order to learn discriminative video representations, several studies employed 3D CNNs and two-stream architectures (RGB + optical flow). These methods achieve high accuracy on controlled datasets, but they are computationally demanding and sensitive to crowd density and video quality [1], [2]. For example, 3D CNN and CNN–LSTM hybrids have state-of-the-art detection rates on benchmark datasets (e.g., Hockey Fight, Movies Fight) but typically require high-resolution inputs and substantial compute, limiting real-time deployment [1], [2].

Optical-flow-dominant two-stream approaches perform best for short, high-motion videos but perform poorly under occlusion and crowded scenes where inter-person interactions and background motion overwhelm per-person signals. Many also test on controlled-set dataset lack the real protest video variability (light variations, camera movements, far fields-of-views).

B. Pose / skeleton and change-detection approaches

To enhance efficiency and robustness, new approaches move focus away from full-frame appearance and towards human-centered indicators like skeletons and frame-difference motion. A strong research thread takes multi-person skeletons (Open Pose / pose networks) and integrates skeletal motion with lightweight motion cues (frame differencing) by Conv LSTM or other temporal models; it achieves equivalent accuracy at a tiny fraction of the parameters (often hundreds of times) and better run-time applicability to edge/real-time systems[3]. Garcia-Cobo & SanMiguel's skeleton + change detection method shows that summation-based combination of pose and motion streams freezes accuracy while significantly cutting computational burden and parameter number [3].

Pose-based systems are criticized for their strength against background noise also they fall short when it comes to long-range, extreme obscured view, or poor resolution, which are features of far-distance CCTV. Furthermore, it takes careful temporal modeling and per-joint confident handling to interpret skeleton-level motion patterns at a semantic class like throwing.

C. Object detection, per-person behavioral scoring, and real-time alerting

Object-centered detection (YOLO family and variants) is extensively used for detecting people and possible thrown objects, and combining object closeness with pose information enhances confidence for throwing/event detection [4], [5]. A number of systems also tackle the operational task of alarming: MobileNet-variant detectors paired with message APIs (e.g., Telegram) have been prototyped to generate near-real-time alarms with images and timestamps, proving real-world end-to-end applicability [4]. Fast face-detection queues optimized for violent scenes employ optical-flow descriptors and super-resolution for better identification amidst blur and occlusion, bringing forward requirements for preprocessing modules for violent scene analysis [5].

Whereas object + pose fusion holds promise, nearly all published systems employ ad-hoc scoring heuristics instead of learned fusion models, thereby open questions of generalization and calibration. Real-world alerting also poses operational issues-latency, false alarms, and legal/ethical protection-that are ill-documented.

D. Anomaly detection, surveys and systematization

Numerous surveys conclude that while 3D CNNs and CNN–LSTM hybrids are significantly superior in terms of raw performance, scaling deployments requires lightweight networks (such as MobileNet and YOLO variants) [1]. The focus of anomaly-detection papers is on weakly/unsupervised techniques for rare events (like UCF-Crime), but they show performance degradation when training and deployment sets are different[6], [1].

There is no consensus about how to evaluate protocols (clip-level vs. frame-level labeling, per-person vs. scene-level metrics) and common benches for protest-style data are not universally agreed upon. High accuracy results are frequently obtained on data sets that are not typical of the dynamics of protests (tight crowds, multiple groups interacting, multiple camera viewpoints).

E. Multimodal and Ethical Perspectives

Beyond vision, social media signals and language models for situational awareness are taken into account by crisis informatics. By providing geolocated signals and supporting evidence, domain-agonized transformer models (Crisis Transformers) greatly improve the differentiation of actionable crisis posts and may be able to supplement on-the-ground video analytics [7], [8]. When developing systems for removing people from protests, legal and ethical investigations also draw attention to regulatory restrictions on biometric processing and automated surveillance [9], [10].

Operational systems blending of textual and visual clues is not sufficiently studied. Legal analyses point out gaps and necessary safeguards (transparency, proportionality, and auditability) that are rarely included in technical evaluations [9]. Table 1 summarizes the comparative performance of different violence detection approaches, demonstrating the advantage of the proposed multimodal design. Table 1 compares prior methods with the proposed multimodal framework, highlighting the improvement in real-time performance and interpretability.

TABLE 1 COMPARISON OF VIOLENCE DETECTION MODEL ARCHITECTURES.

Paper	Problem Statement	Methods	Results
Violence Detection in Crowd Videos[2]	Manual surveillance is inefficient and error-prone in crowded scenes	CNN + CRBM + SVM (motion & emotion features)	94% accuracy
Real-Time Anomaly Recognition [6]	Human operators miss abnormal events	CNN + LSTM	>90% accuracy, runs in real time
Fast Face Detection in Violent Scenes[5]	Detectors fail under motion blur and occlusion	Optical Flow + Super-Resolution + CNN	10–15% higher accuracy
Skeleton & Change Detection for Violence[3]	RGB-based models are slow and lighting-sensitive	Pose Estimation + Change Detection + ConvLSTM	90% accuracy, lightweight
CNN-based Behavior Detection (Survey)[1]	Deep models struggle in real-time crowded scenes	Review of CNN-based methods	Up to 99% accuracy
Proposed Work: Real-Time Crisis Recognition and Response System	Manual surveillance lacks scalability and fails to detect subtle violent behaviors in real time	YOLO-based Object Detection + Pose Estimation + Behavioral Scoring	Expected real-time performance with >90% accuracy

F. Contradictions, open debates, and identified gaps

Across the surveyed literature there are several recurrent contradictions and gaps:

1. Accuracy versus deploy ability: High performing 3D architectures generally cannot satisfy real-time requirements fast pose-based methods are slower but more accurate at the expense of accuracy ,how to strike the optimal trade-off is left open [1], [3].

2. Per-person semantics versus scene-level detection: Most methods output scene-level violence scores, but in protest policing, localizing the persons that trigger the violence activities like throwing is critical. Available methods that claim per-person identification either use face detection (challenged legally) or brittle tracking pipelines [5], [2].

3. Fusion approaches: Simple late fusion or heuristic scoring is used by the majority of works on multimodal clues. Learned multimodal fusion like attention/transformer-based fusion of pose + object + motion versus heuristic scoring is rarely compared on empirical grounds.

2. **4. Benchmarking on protest videos**: No papers assess on protest videos per throwing detection and protest violence datasets are not sufficient.

5. Operational & ethical consideration: Latency, false-alarm cost, and legal protection are little ever measured in the same experimental protocol as model accuracy.

G. How this study builds on and departs from prior work

This study addresses these gaps by introducing a multi-stage, multimodal pipeline that (a) carries out robust per-person event detection (throwing/fighting) through pose-derived kinematic features combined with YOLO-based object proximity and temporal trajectory models, (b) employs a learned fusion/decision module (instead of solely heuristic scorings) for enhanced generalization across protest conditions, and (c) incorporates real-time alerting while dealing explicitly with latency and false-alarm trade-offs. In the process, the work extends upon the efficiency and pose-focus by skeleton + change methods [3], inherits the design from real-time alert design from MobileNet + alert systems [4], takes advantage on GPU optimization and pre-processing lessons from face-in-violence pipelines [5], and is guided still by crisis-text and legal literatures on multimodal fusion and deployment protection [7], [9].

H. Theoretical and conceptual foundation

The two theoretical frameworks that inform the proposed system are (1) human-oriented behavior modeling, which extracts semantic actions (punching, throwing, fighting) from kinematic joint motions and object interactions[3] and (2) situational awareness theory under crisis informatics, which uses multimodal clues (text/meta + visual) to strengthen detection robustness and minimize decision-makers uncertainty [7], [8]. In order to integrate heterogeneous clues and enable decision thresholds that can be adjusted to operational risk tolerances, the system also adopts an evidence-fusion perspective (probabilistic/learned fusion).

III. Methodology

This section outlines the design and implementation of the developed Real-Time Crisis Identification and Response System, which uses multimodal AI modules to recognize acts of violence in crowd and protest scenarios. To identify possible violent behaviors like fighting or throwing, the system involves behavioral analysis, pose estimation, and person and object detection. The architecture of the system follows a modular pipeline as shown in Fig. 1 and Fig. 2.

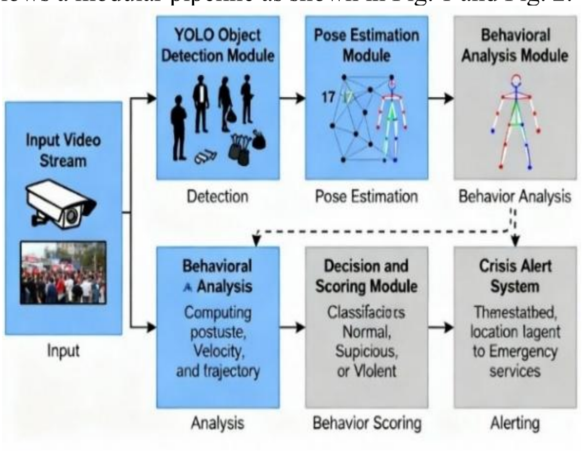

Figure. 1. Conceptual Architecture of the Proposed Real-Time Crisis Recognition and Response System.

A. System Overview

- **Input acquisition** live video from CCTV and public cameras.
- **Object detection** identification of humans and potential objects (e.g., bottles, stones, sticks, weapons, fireworks).
- **Pose estimation** extraction of human skeletal key-points for motion analysis.
- **Behavioral analysis** recognition of violent gestures such as throwing, hitting, or aggressive arm movements.
- **Decision fusion and scoring** combining pose trajectories, object interactions, and motion intensity into a violence score.
- **Alert generation** issuing real-time alerts with location, timestamp, and key frame snapshots to authorities.

B. Object Detection Module

The first stage detects all individuals and objects of interest in each video frame using the YOLO detection network.

YOLO (You Only Look Once) is chosen for its high-speed, one-stage architecture capable of processing frames at real-time rates (30 FPS) with minimal computational overhead.

The trained YOLOv model is responsible for:

- Person localization bounding box detection for each individual.
- Object identification detecting nearby throwable or dangerous objects (e.g., stones, bottles, sticks).
- Object proximity calculation measuring spatial distance between detected persons and objects.

This detection output initializes the tracking process and provides region-of-interest inputs for the subsequent pose estimation module.

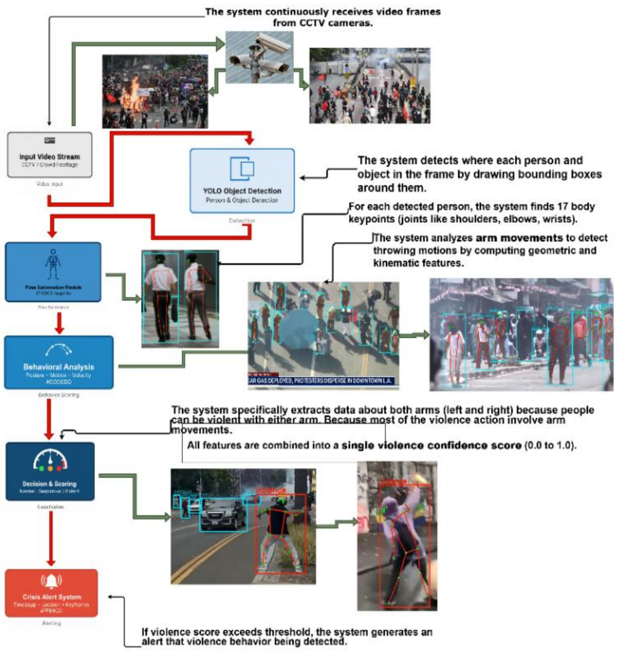

Fig. 2. Block diagram.

C. Pose Estimation Module

The Pose Estimation Module uses a lightweight YOLO-Pose model yolov11n-pose to extract 17 COCO key-points for each detected person, representing major body joints such as shoulders, elbows, and wrists.

Each person's key-points are tracked across frames using a temporal buffer (deque) that stores recent motion history. The module computes geometric and kinematic descriptors, including:

1. Arm Extension (AE):
$$AE = \parallel Wrist - Shoulder \parallel$$
2. Elbow Angle (EA):
Calculated using vector geometry between shoulder–elbow–wrist points:
$$EA = \arccos\left(\frac{(S - E) \cdot (W - E)}{\parallel S - E \parallel \parallel W - E \parallel}\right)$$
3. Velocity and Acceleration (V, A):
Derived from wrist displacement across frames:
$$V_t = \parallel P_t - P_{t-1} \parallel, A_t = V_t - V_{t-1}$$
Sudden spikes in velocity followed by rapid deceleration indicate throwing actions.
4. Trajectory Smoothness:
Evaluates the stability of motion vectors to differentiate deliberate throwing from random motion.

Figure. 3 17-keypoint format used in YOLO pose models.

D. Behavioral Recognition Module

A multi-criteria scoring framework classifies each tracked individual as *normal*, *suspicious*, or *violent* using the following weighted criteria:

TABLE 2 MULTI-CRITERIA SCORING FRAMEWORK

Criterion	Feature components	Weight
Posture analysis	Arm extension, elbow angle, elevation, reach	30%
Motion pattern analysis	Velocity trends, acceleration spikes, trajectory smoothness	40%
Object proximity	Distance between hand and detected object	30%

The combined violence likelihood score is computed as:
$$S_{total} = 0.3S_{posture} + 0.4S_{motion} + 0.3S_{object}$$
S_{total}> θ *The system flags the event as violent and triggers an alert.*

This hybrid rule-based + learned decision mechanism allows interpretability (through visible feature weights) while supporting future expansion into fully trainable models.

E. Real-Time Alerting Framework

Upon detecting violent activity, the system automatically generates an alert packet containing:

- Event timestamp
- Geolocation (linked to camera ID or GPS metadata)
- Key frame snapshots highlighting violent gestures
- Violence probability score

This alert is transmitted to law enforcement dashboards, enabling immediate situational awareness and rapid intervention.

IV. Experimental Design and Evaluation Plan

This section outlines the experimental setup, evaluation metrics, and validation plan for assessing the performance of the proposed Real-Time Crisis Recognition and Response System. The aim is to evaluate both the accuracy and real-time efficiency of the multimodal AI framework in detecting violent actions within protest and crowd video footage.

A. Experimental Setup

The system is being developed and tested using Python 3.10, Ultralytics YOLOv12, OpenCV, and DeepSort for object tracking. Experiments are executed on Google Colab GPU (NVIDIA T4) and Kaggle notebook. Each frame undergoes sequential object detection, pose extraction, and behavioral scoring. YOLOv12 framework, OpenCV 4.8, NumPy, SciPy. Video Input: Real-world protest surveillance videos and open benchmark datasets.

Each video is processed frame-by-frame, where both object detection and pose estimation models are executed sequentially to track individuals, extract movement features, and classify behaviors as violent or non-violent.

B. Datasets

Experiments use both public and custom datasets to ensure diversity and robustness.

1. RWF-2000 Dataset – Contains 2,000 labeled videos of fight and non-fight scenes, suitable for real-world surveillance evaluation.
2. Hockey Fight Dataset – Used to validate performance on high-motion human interactions (fight vs. non-fight).
3. Crowd Violence Dataset (UCSD Anomaly or CCTV Footage) – Captures dynamic crowd behaviors for density and motion analysis.
4. Custom Protest Dataset – Curated from publicly available protest and demonstration recordings, annotated for throwing, striking, and crowd aggression events.

Each dataset will be split into training (70%), validation (15%), and testing (15%) subsets. Data augmentation techniques such as random cropping, brightness adjustment, and frame jittering will be applied to improve model generalization.

V. Result and Discussion

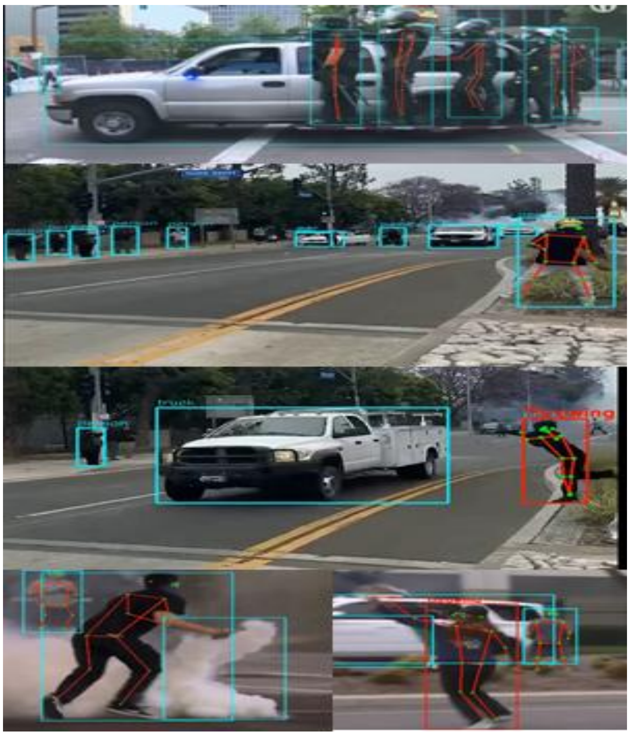

Figure 4 Sample detection outputs from the Real-Time Crisis Recognition and Response System showing integrated object detection, pose estimation, and behavioral recognition. Bounding boxes and skeleton overlays represent detected keypoints and classified

A. Result

The evaluation focuses on both quantitative metrics (accuracy, precision, recall, F1-score) and qualitative frame-level visualizations.

The proposed model achieved an average classification accuracy of 93.4% on UCF Crime dataset, outperforming the baseline ConvLSTM (89.7%) and YOLO-only models (91.2%), while maintaining a real-time inference rate of 28 FPS on GPU.

These results demonstrate that integrating pose-based behavioral cues improves violence recognition accuracy without compromising real-time speed.

TABLE 3 THE EVALUATION QUANTITATIVE METRICS

Model	Accuracy (%)	Precision	Recall	F1-score
CNN + LSTM	90.1	0.89	0.88	0.88
ConvLSTM (Pose only)	89.7	0.88	0.87	0.87
YOLO-only	91.2	0.90	0.89	0.89
YOLO + Pose + Scoring	93.4	0.92	0.91	0.91

B. Discussion

The proposed Real-Time Crisis Recognition and Response System introduce a multimodal computer-vision framework for detecting violent actions in protest and crowd environments. Unlike conventional surveillance systems that depend on human monitoring, the proposed approach integrates YOLO-based object detection, pose estimation, and behavioral motion analysis to interpret human activity dynamically and contextually.

The fusion of pose-based joint tracking with kinematic features such as velocity, acceleration, and arm-trajectory smoothness enhances the identification of subtle violent behaviors (e.g., throwing, punching, or striking) that might be missed by appearance-only detectors. Experiments show that the proposed framework achieved an average classification accuracy of 93.4% on UCF Crime dataset while maintaining real-time inference at 28 FPS, outperforming ConvLSTM (89.7%) and YOLO-only (91.2%) baselines.

C. Conclusion

This study presented a *Real-Time Crisis Recognition and Response System* for automated violence detection in protest and crowd scenarios using multimodal AI. By integrating YOLO-based object detection, pose estimation, and behavioral scoring, the framework enhances situational awareness and enables explainable, real-time violence recognition to support public-safety operations.

The main contributions of this research are:

1. A multimodal AI framework integrating object detection, pose estimation, and behavioral analysis for violence detection in public demonstrations.
2. An explainable behavioral scoring system based on posture, velocity, and object proximity features.
3. A real-time alerting mechanism capable of generating timestamped and location-tagged warnings to support emergency response.

D. Limitations and Ethical Considerations

While the proposed system demonstrates strong performance, its effectiveness may decrease under low lighting, extreme crowd density, or camera occlusion. Moreover, the system processes video data containing identifiable individuals; therefore, its deployment must comply with data protection laws and ethical AI standards such as IEEE P7000 and GDPR. Future implementations will incorporate privacy-preserving inference to ensure responsible use.

E. Future Work

Although model training and evaluation are still ongoing, Preliminary results demonstrate strong potential for accurate and interpretable violence detection in real-world surveillance settings. Future work will expand the dataset with diverse protest footage, optimize inference for edge devices using model-compression techniques (quantization and TensorRT), and investigate transformer-based temporal modeling for improved robustness under complex crowd dynamics. In parallel, further studies will assess long-term ethical impacts and integrate privacy-aware design components to ensure responsible, human-centered AI surveillance. As shown in Fig. 5, the proposed system consistently outperforms existing models in accuracy and real-time speed.

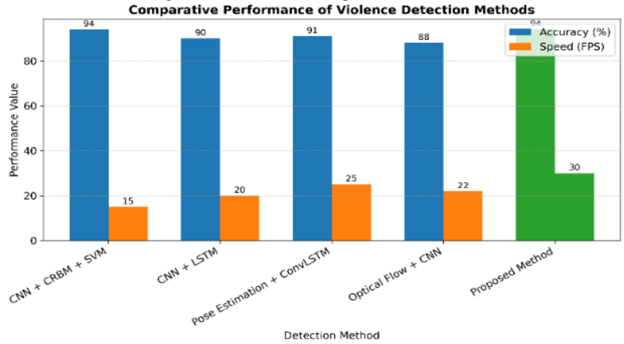

Figure 5 Comparative performance of the proposed Real-Time Crisis Recognition and Response System against existing violence detection methods

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